FEDERATED LEARNING-BASED PNEUMONIA DETECTION IN SMART HEALTHCARE: A COMPARATIVE STUDY

A PROJECT REPORT

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Abstract

This project presents a privacy-preserving and efficient approach for pneumonia detection using chest X-ray images by integrating deep learning with federated learning. Pneumonia is a critical respiratory disease that requires early and accurate diagnosis; however, traditional methods relying on manual interpretation by radiologists are time-consuming, error-prone, and not scalable in high-demand or resource-limited settings.

To address these challenges, the proposed system utilizes the EfficientNet-B4 deep learning architecture for image classification, combined with a Federated Learning framework that enables multiple clients (simulated as hospitals) to collaboratively train a global model without sharing sensitive patient data. Each client performs local training and shares only model parameters with a central server, ensuring data privacy and security.

The system follows a structured pipeline including data preprocessing, augmentation, distributed training, federated aggregation using the FedAvg algorithm, and performance evaluation. The model achieved an accuracy of approximately 90.71% with strong recall and an AUC score of 0.964, demonstrating high effectiveness in distinguishing between pneumonia and normal cases.

The results indicate that the proposed approach not only improves diagnostic accuracy but also ensures data privacy, scalability, and real-world applicability. This work highlights the potential of combining deep learning with federated learning to develop secure and reliable intelligent healthcare systems

CHAPTER 1

INTRODUCTION

1.1 Introduction to Project

Pneumonia is a severe respiratory infection that affects millions of people worldwide and remains a major cause of illness and death, particularly among children, elderly individuals, and patients with weakened immune systems. The disease causes inflammation in the air sacs of the lungs, leading to symptoms such as cough, fever, chest pain, and difficulty in breathing. Early and accurate diagnosis of pneumonia is essential to ensure timely treatment and to reduce complications and mortality rates.

Chest X-ray imaging is one of the most commonly used diagnostic tools for detecting pneumonia due to its affordability and wide availability in healthcare systems. However, the interpretation of X-ray images requires experienced radiologists and can be time-consuming, especially in hospitals with a large number of patients.
In addition, human interpretation may sometimes lead to inconsistencies or errors, which can affect the reliability of the diagnosis. These challenges highlight the need for an automated system that can assist healthcare professionals in detecting pneumonia more efficiently and accurately.

With the advancement of artificial intelligence and deep learning, significant improvements have been made in medical image analysis. Convolutional Neural Networks (CNNs) have proven to be highly effective in extracting meaningful patterns from images and performing classification tasks. In this project, a deep learning-based approach is used to detect pneumonia from chest X-ray images using the EfficientNet-B4 architecture, which is known for its high accuracy and efficiency in image classification.

To address data privacy concerns in healthcare, this project adopts a Federated Learning approach. Instead of collecting all data in a centralized location, the dataset is distributed across multiple clients (simulated as hospitals), and the model is trained collaboratively. Each client trains the model locally on its own data and shares only the model parameters with a central server for aggregation. This ensures that sensitive patient data remains private while still benefiting from collective learning.

The proposed system follows a structured workflow that includes data preprocessing, model training, federated aggregation, and performance evaluation. The model achieved a high accuracy of approximately

88.94%, demonstrating its effectiveness in pneumonia detection. This approach not only improves diagnostic accuracy but also ensures data privacy, making it suitable for real-world healthcare applications.

1.2 Problem Statement

Pneumonia is a life-threatening respiratory disease that requires early and accurate diagnosis for effective treatment. However, in most healthcare systems, diagnosis still relies heavily on the manual interpretation of chest X-ray images by radiologists. This process is time-consuming, requires high expertise, and may lead to variability in results due to human error. In busy hospitals with a large number of patients, delays in diagnosis can negatively impact patient outcomes.

The problem becomes more critical in rural and resource-limited areas where access to skilled radiologists is limited. Patients in such regions often experience delays in diagnosis and treatment, increasing the risk of severe complications. Additionally, the increasing volume of medical data further adds pressure on healthcare professionals, making it difficult to maintain both speed and accuracy in diagnostic procedures.

Another major challenge in existing deep learning-based systems is the dependency on centralized data. Most traditional models require all medical data to be collected and stored in a single location for training. In the healthcare domain, this raises serious concerns regarding data privacy, security, and regulatory compliance, as patient data is highly sensitive and cannot be freely shared across institutions.

Furthermore, many existing models function as “black-box” systems, where predictions are generated without providing clear explanations. This lack of transparency reduces trust among medical professionals, limiting the adoption of such systems in real-world clinical environments.

To address these challenges, this project proposes a Federated Learning-based pneumonia detection system using chest X-ray images. The system allows multiple healthcare institutions (clients) to collaboratively train a deep learning model without sharing their raw data. Each client

trains the model locally and shares only model parameters with a central server, ensuring data privacy and security.

The proposed approach aims to provide a reliable, efficient, and privacy-preserving solution for pneumonia detection. By combining deep learning with federated learning, the system reduces dependency on centralized data, improves diagnostic accuracy, and supports healthcare professionals in making faster and more consistent decisions.

1.3 Motivation

The motivation for this project arises from the growing need for accurate, efficient, and privacy-preserving healthcare solutions, particularly in the field of medical image analysis. Pneumonia continues to be one of the leading causes of death worldwide, especially among children, elderly individuals, and patients with weak immune systems. Early detection plays a critical role in reducing complications and improving survival rates.

In many hospitals, especially those with high patient volumes, radiologists are required to analyze a large number of X-ray images within limited time. This increases the chances of fatigue and human error, which may lead to incorrect or delayed diagnosis. An automated system powered by deep learning can assist healthcare professionals by providing faster and more consistent results, thereby improving the overall efficiency of the diagnostic process.

Another important motivation is the issue of data privacy in healthcare. Medical data is highly sensitive, and sharing patient information across institutions is often restricted due to legal and ethical concerns. Traditional centralized machine learning approaches require data to be collected in a single location, which increases the risk of data breaches. Federated Learning offers a solution to this problem by enabling collaborative model training without sharing raw data, ensuring that patient information remains secure and private.

The project is also motivated by the need to improve healthcare accessibility in rural and underdeveloped regions. In many such areas, access to experienced radiologists and advanced medical facilities is limited. An

AI-based system can be deployed to assist healthcare providers in these regions, enabling early detection of pneumonia and timely medical intervention.

Furthermore, this project aims to contribute to the advancement of intelligent healthcare systems that combine artificial intelligence with real-world medical applications. By integrating EfficientNet-B4 with Federated

Learning, the system not only achieves high accuracy but also ensures scalability and privacy. The overall goal is to develop a reliable and practical solution that can support healthcare professionals, improve patient outcomes, and promote the adoption of AI in medical diagnosis.

1.4 Sustainable Development Goal of the Project

The proposed project aligns with the United Nations Sustainable Development Goals (SDGs), particularly focusing on improving healthcare systems and promoting the use of advanced technologies for societal benefit. The primary contribution of this project is towards SDG 3: Good Health and Well-being, which aims to ensure healthy lives and promote well-being for people of all ages. By enabling early and accurate detection of pneumonia, the system helps in reducing the risk of severe illness and mortality.

Timely diagnosis allows healthcare professionals to initiate appropriate treatment at an early stage, thereby improving patient outcomes. The use of artificial intelligence in medical diagnosis also enhances the efficiency of healthcare services by reducing the workload on radiologists and minimizing diagnostic delays. The project also contributes to SDG 9: Industry, Innovation, and Infrastructure by incorporating advanced technologies such as deep learning and federated learning. These technologies demonstrate how modern AI solutions can be applied to solve real-world healthcare challenges.

The federated learning approach, in particular, promotes secure and collaborative innovation by enabling multiple institutions to train models without sharing sensitive data. In addition, the system supports SDG 10: Reduced Inequalities by improving access to diagnostic tools in remote and underserved regions. Since the model can be deployed in decentralized environments, healthcare providers in areas with limited resources can benefit from AI-assisted diagnosis without requiring extensive infrastructure or specialized expertise. Overall, this project contributes to sustainable development by combining healthcare improvement, technological innovation, and data privacy. It highlights the potential of intelligent systems in creating a more accessible, efficient, and secure healthcare ecosystem.

CHAPTER 2

LITERATURE SURVEY

2.1. Overview of the Research Area

The field of medical image analysis has experienced significant growth in recent years due to advancements in artificial intelligence and deep learning techniques. These technologies have transformed the way diseases are detected and diagnosed, particularly in areas where visual interpretation of medical images plays a crucial role.

Among various imaging techniques, chest X-ray imaging is widely used for diagnosing respiratory diseases such as pneumonia because it is cost-effective, easily accessible, and provides quick insights into lung conditions.

Despite its advantages, manual analysis of chest X-ray images remains a challenging task. It requires skilled radiologists to carefully examine the images and identify subtle patterns associated with diseases. In healthcare environments with a high patient load, this process can become time-consuming and may lead to delays in diagnosis. Additionally, variations in human interpretation can result in inconsistent or inaccurate diagnoses, which can affect patient treatment and outcomes. These challenges have encouraged the development of automated systems that can assist in medical image analysis.

Deep learning, particularly Convolutional Neural Networks (CNNs), has emerged as a powerful tool for image classification and feature extraction. CNNs are capable of automatically learning complex patterns from images without the need for manual feature engineering. Advanced architectures such as ResNet, DenseNet, and EfficientNet have significantly improved performance in medical image classification tasks, including pneumonia detection. Among these, EfficientNet is known for its ability to achieve high accuracy while maintaining computational efficiency.

In addition to accuracy, data privacy has become an important concern in healthcare applications. Traditional machine learning approaches rely on centralized data, where all data is collected and processed in a single location. However, sharing medical data across institutions raises serious privacy and security issues. To address this challenge, Federated Learning has been introduced as a distributed learning approach that allows multiple clients to collaboratively train a model without sharing raw data. This approach ensures data privacy while enabling effective learning from decentralized datasets.

Overall, current research focuses on developing intelligent systems that combine deep learning with privacy- preserving techniques to improve diagnostic accuracy and efficiency. The integration of Federated Learning with advanced deep learning models represents a promising direction for building secure and reliable healthcare applications.

2.2. Existing Models and Frameworks

Several research studies have explored the use of deep learning techniques for pneumonia detection using chest

X-ray images. In the early stages, basic Convolutional Neural Networks (CNNs) were widely

used for image classification

tasks. These models demonstrated the ability to automatically learn features from medical

images, eliminating the need for manual feature extraction. However, their performance was limited when trained on small datasets and required significant computational resources.

To overcome these limitations, transfer learning became a popular approach in medical image analysis. Pre- trained models such as VGGNet, ResNet, and DenseNet, which are trained on large datasets like ImageNet, are fine-tuned for specific tasks such as pneumonia detection. This approach improves accuracy and reduces training time. Among these models, DenseNet-based architectures have shown strong performance in detecting pneumonia and, in some cases, have achieved results comparable to expert radiologists.

Recent advancements introduced more efficient architectures such as EfficientNet, which uses a balanced scaling method to optimize model depth, width, and resolution. EfficientNet models provide high accuracy while maintaining computational efficiency, making them suitable for medical imaging tasks. In addition, lightweight models such as MobileNet and small CNN architectures have been developed for deployment in resource-constrained environments, such as mobile devices and edge systems.

Another important development in this field is the use of ensemble learning techniques. Instead of relying on a single model, ensemble methods combine multiple models to improve prediction accuracy and robustness. By integrating the strengths of different architectures, ensemble approaches can capture diverse features and reduce overfitting, leading to better generalization.

Despite these advancements, most existing models are based on centralized learning, where all training data is stored in a single location. This approach raises concerns regarding data privacy and security, especially in healthcare applications where patient data is sensitive. To address this issue, Federated Learning has emerged as a promising framework that enables collaborative training across multiple institutions without sharing raw data. In this approach, each client trains the model locally and shares only model updates with a central server, ensuring privacy while maintaining performance.

Overall, existing models and frameworks have significantly improved pneumonia detection accuracy.

However, integrating advanced architectures like EfficientNet with privacy-preserving techniques such as

Federated Learning offers a more secure, scalable, and practical solution for real-world healthcare applications.

2.3. Limitations Identified from Literature Survey (Research Gaps)

Despite the significant advancements in deep learning-based approaches for pneumonia detection, several limitations still exist in current research and systems. One of the major challenges is the availability of high- quality and well-balanced datasets. Many publicly available medical datasets suffer from class imbalance, where one class (such as pneumonia) may have significantly more samples than the other (normal cases). This imbalance can lead to biased predictions, reducing the reliability and overall performance of the model. Another imports limitation is the dependency on centralized data storage. Most existing deep learning models require all training data to be collected and stored in a single location. In the healthcare domain, this raises serious concerns regarding data privacy, security, and regulatory compliance. Sharing sensitive patient data across multiple institutions is often restricted, making centralized approaches less practical for real-world applications. Lack of interpretability is also a significant issue in many deep learning models. These models often function as “black-box” systems, where predictions are generated without clear explanations. In medical applications, this lack of transparency reduces trust among healthcare professionals, as they need to understand the reasoning behind the model’s decisions before relying on them for diagnosis. Overfitting is another common challenge, especially when models are trained on limited datasets. In such cases, the model may perform well on training data but fail to generalize effectively to new, unseen data. This reduces the model’s robustness and limits its practical usability in clinical environments. Additionally, many advanced deep learning models require high computational resources for training and deployment. This makes it difficult to use such systems in resource-constrained environments, such as rural healthcare centers or small clinics with limited infrastructure. Finally, most existing systems focus only on binary classification, distinguishing between normal and pneumonia cases. However, real-world medical diagnosis often involves identifying different types and severity levels of diseases. The lack of multi-class classification limits the scope and applicability of these systems. These limitations highlight the need for a more efficient, scalable, and privacy-preserving solution. The integration of Federated Learning with advanced deep learning models addresses many of these challenges by enabling secure distributed training, improving generalization, and making the system more suitable for real- world healthcare applications.

2.4 Research objective

The primary objective of this project is to develop an efficient and reliable system for pneumonia detection using chest X-ray images by leveraging deep learning techniques. The system aims to accurately classify medical images into normal and pneumonia categories, assisting healthcare professionals in making faster and more informed diagnostic decisions.

Another important objective is to integrate Federated Learning into the model development process. This approach enables multiple distributed clients (such as hospitals) to collaboratively train a global model without sharing their raw data. By doing so, the system ensures data privacy and security, which are critical requirements in healthcare applications.

The project also aims to utilize the EfficientNet-B4 architecture to achieve high classification accuracy while maintaining computational efficiency. EfficientNet is selected due to its ability to balance model performance and resource utilization, making it suitable for medical image analysis tasks.

In addition, the project focuses on improving model generalization by applying preprocessing and data augmentation techniques such as resizing, normalization, rotation, and flipping. These techniques help the model learn robust features and reduce the risk of overfitting.

Another objective is to evaluate the performance of the proposed system using standard metrics such as accuracy, precision, recall, F1-score, and AUC. These metrics provide a comprehensive understanding of the model's effectiveness and reliability in detecting pneumonia.

Finally, the project aims to develop a scalable and practical solution that can be applied in real-world healthcare environments. By combining deep learning with federated learning, the system is designed to improve diagnostic accuracy, ensure data privacy, and support healthcare professionals in delivering better patient care

2.5 Product user stories with Desired outcomes)

2.5 Product Backlog (Key User Stories with Desired Outcomes)

The product backlog defines the key functionalities of the federated learning system in the form of user stories.

S.No	User Story	Desired Outcome
US1	As a **hospital node**, I want to train a local EfficientNet-B4 model on my private chest X-ray data so that patient data never leaves the hospital	Ensure data privacy and local training
US2	As a **central server**, I want to receive encrypted model weights from all hospital nodes so that I can aggregate them without accessing raw data	Secure weight aggregation
US3	As a **central server**, I want to apply the FedAvg algorithm to combine local model weights so that a globally optimized model is created	Generate optimized global model
US4	As a **hospital node**, I want to receive the updated global model after each federated round so that my local model improves over time	Continuous model improvement
US5	As a **user/doctor**, I want to upload a chest X-ray image so that the system can classify it as NORMAL or PNEUMONIA	Accurate pneumonia detection
US6	As a **user/doctor**, I want to view the prediction result with confidence score so that I can make an informed clinical decision	Clear and reliable diagnosis output
US7	As a **admin/developer**, I want to monitor accuracy per federated round so that I can track model convergence	Training performance visibility
US8	As a **admin/developer**, I want to evaluate the global model using accuracy, precision, recall, F1 and AUROC so that model quality is verified	Comprehensive model evaluation
US9	As a **hospital node**, I want communication with the central server to be encrypted so that model updates are protected from interception	Secure federated communication
US10	As a **admin/developer**, I want the system to support 3 hospital nodes simultaneously so that federated training is properly simulated	Multi-client federated simulation

Table 2.5 Detailed user stories

2.6 Plan of Action (Project Road Map)

The project follows a structured workflow to develop the pneumonia detection system using Federated

Learning.

1. Data Collection

Chest X-ray images are collected from a public dataset.

2. Data Preprocessing

Images are resized and normalized for consistency.

3. Data Augmentation

Techniques like rotation and flipping are applied to improve generalization.

4. Dataset Distribution

The dataset is divided into multiple clients (hospitals) for federated learning.

5. Model Development

EfficientNet-B4 is used for image classification.

6. Federated Learning

Clients train the model locally, and a central server aggregates the weights.

7. Model Training

Training is performed over multiple rounds to improve performance.

8. Evaluation

Model performance is measured using accuracy, precision, recall, and F1-score.

9. Results and Saving

Results are visualized, and the best model is saved.

This roadmap ensures efficient and privacy-preserving model develops

CHAPTER 3

SPRINT PLANNING AND EXECTION METHODOLOGY

3.1 SPRINT I (System Design and Initial Development)

3.1.1 Objectives with user stories of Sprint I

The objective of Sprint I is to design and establish the core framework of the pneumonia detection system. This phase focuses on building a structured pipeline that enables image input, preprocessing, and preparation for model training.

The system is designed to accept chest X-ray images from users and process them efficiently.
Preprocessing steps such as resizing and normalization are applied to ensure consistency in the dataset.
This prepares the data for accurate classification using deep learning models.

In this sprint, user requirements are considered to ensure the system is simple and practical. As a user, the system should allow easy upload of X-ray images. As a system, it should preprocess images and prepare them for analysis. As a developer, the system should support smooth data flow for training and evaluation.

The sprint also focuses on preparing the system for Federated Learning by organizing the dataset and defining how data will be distributed across multiple clients. This ensures that the system is ready for collaborative training in later stages.

Overall, Sprint I establishes the foundation of the project by defining system functionality, workflow, and initial setup required for further development.

3.1.2 Functional Document

The functional design of the system defines how different components interact to perform pneumonia detection using chest X-ray images. The system is designed to take input images, preprocess them, and prepare them for model training and evaluation.

The input to the system consists of chest X-ray images collected from a dataset. These images are passed through preprocessing steps such as resizing, normalization, and augmentation to ensure consistency and improve model performance.

The processed data is then distributed across multiple clients (simulated hospitals) as part of the federated learning setup. Each client trains the model locally using its own dataset. The trained model parameters are sent to a central server, where they are aggregated to form a global model.

The system also includes an evaluation component that measures performance using metrics such as accuracy, precision, recall, and F1-score. Additionally, visualization tools such as confusion matrix and ROC curves are used to analyze the results.

Overall, the functional design ensures smooth data flow from input to output, enabling efficient and privacy-preserving pneumonia detection.

3.1.3 Architecture Document

The system architecture is designed for pneumonia detection using Federated Learning. The user interacts with the system through a web or application interface, and input images are sent to a backend server.

The data preprocessing module performs resizing, normalization, and augmentation to prepare the images. The dataset is then distributed across multiple clients (hospitals), ensuring that data remains decentralized.

Each client trains the model locally using EfficientNet, and only model weights are shared with a central federated server. The server aggregates these weights using the Federated Averaging (FedAvg) algorithm to create a global model.

This process is repeated over multiple rounds to improve performance. The final model is evaluated using metrics such as accuracy, precision, recall, and F1-score, and results are displayed through a dashboard.

This architecture ensures efficient training, high accuracy, and data privacy in healthcare applications.

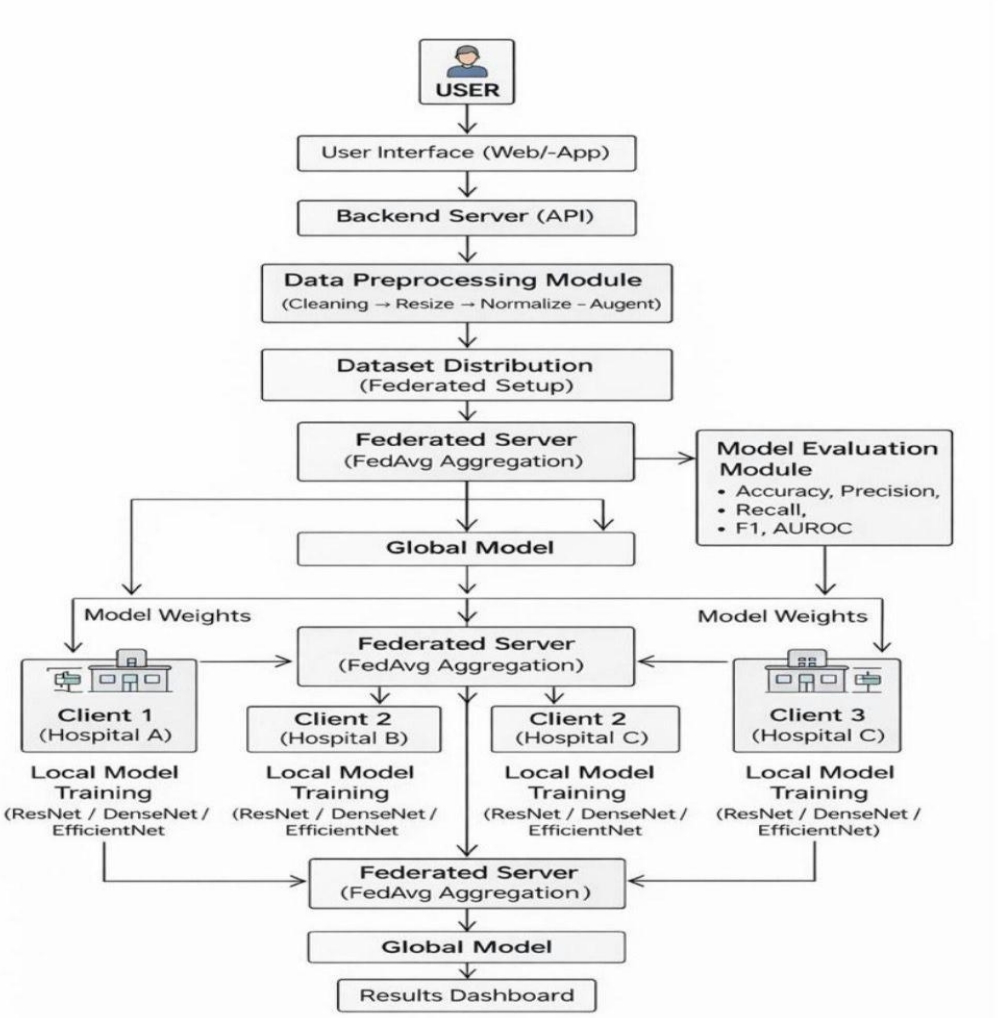


Figure 1:Architecture Diagram

3.1.4 Outcome of objectives/ Result Analysis

The objectives of Sprint I were achieved by setting up the system design and data preprocessing pipeline. The dataset was successfully prepared and distributed across multiple clients for federated learning.

The workflow from data input to distribution was established, ensuring smooth data handling. This sprint provided a strong foundation for model training and further development.

3.1.5 Sprint Retrospective

Sprint I was completed successfully with all planned objectives achieved. The federated learning framework was designed, the dataset was prepared and partitioned into three hospital nodes, and the EfficientNet-B4 model was trained using the FedAvg aggregation algorithm. Minor challenges related to dataset path configuration and GPU memory constraints were resolved during development.

The sprint highlighted the importance of structured federated architecture design and proper checkpoint management. The team gained a clear understanding of federated learning workflows, client-server communication, and model aggregation, which will support further development and optimization in upcoming sprints.

Sprint Retrospective			
What went well	**What went poorly**	**What ideas do you have**	**How should we take action**
Successfully collected and organized chest X-ray dataset (5,863 images) Federated split into 3 hospital nodes completed EfficientNet-B4 pretrained model loaded successfully FedAvg aggregation implemented and verified Local training on each client executed without errors Team coordination and task distribution were effective	Validation set had only 16 images causing unstable validation metrics Training time per round was high on free GPU Dataset path errors caused delays in Kaggle setup Memory constraints limited batch size options Class imbalance between NORMAL and PNEUMONIA affected early rounds	Use train set portion as validation to fix small val set issue Add more federated rounds for better convergence Apply class weighting to handle imbalance Explore fine-tuning more layers for accuracy improvement Add DenseNet and ResNet as comparison baselines	Increase rounds from 3 to 10 for better global model Use Kaggle GPU T4 x2 for faster training Save checkpoints after every round to prevent data loss Apply data augmentation to balance class distribution Validate dataset path before running full training
EfficientNet-B4 federated training achieved 90.71% accuracy	Small validation set caused metric instability	Use larger validation split for future sprints	Apply stratified sampling for balanced validation
FedAvg aggregation worked correctly across 3 clients	GPU runtime limits on free Colab/Kaggle	Automate checkpoint saving after every round	Use Kaggle T4 x2 GPU for faster execution
Privacy preserved — no raw data shared between nodes	Accuracy fluctuated across rounds 4–5	Increase local epochs per round for stability	Use learning rate scheduler for stable convergence

3.2 SPRINT II (Testing and Evaluation)

3.2.1 Objectives with User Stories of Sprint II

The objective of Sprint II is to train, test, and evaluate the pneumonia detection model using the federated learning approach. This phase focuses on improving model performance and validating its effectiveness.

As a user, the system should provide accurate predictions for chest X-ray images. As a developer, the system should support efficient federated training and evaluation. As a system, it should measure performance using metrics such as accuracy, precision, recall, and F1-score.

The sprint also includes evaluating the global model after aggregation and analyzing results using tools like confusion matrix and ROC curve.

Overall, Sprint II ensures proper testing, evaluation, and validation of the model for reliable performance.

3.2.2 Functional Document

The testing phase evaluates system functionality using predefined test cases. According to the functional test case document, various scenarios are tested, including valid image upload, invalid file handling, federated training rounds, FedAvg aggregation, model prediction accuracy, and privacy preservation. The system successfully uploads valid chest X-ray images and rejects unsupported formats. The EfficientNet-B4 federated model correctly classifies pneumonia and normal cases with 90.71% accuracy. The FedAvg aggregation algorithm successfully combines model weights from 3 hospital nodes without transferring any raw patient data, demonstrating reliable privacy-preserving performance.

Feature	Test Case	Steps to execute test case	Expected Output	Actual Output	Status	More Information
Federated Learning	Valid Chest X-Ray Upload	1. Open the application. 2. Click upload button. 3. Select a valid chest X- ray image (.jpg/.png). 4. Click submit.	1. Image should be uploaded successfully. 2. System should display uploaded image.	1. Image uploaded successfully. 2. Image displayed correctly.	Pass	No error message displayed. Image format supported (.jpg/.png).
Federated Learning	Invalid File Upload	1. Open application. 2. Upload non-image file (.pdf/.txt). 3. Click submit.	1. System should reject file. 2. Error message displayed.	1. File rejected successfully. 2. Error message shown.	Pass	Input validation working correctly.
FL Training	Federated Round Execution	1. Initialize central server. 2. Broadcast global model to 3 hospital clients. 3. Each client trains locally. 4. Send weights to server.	1. Each client trains successfully. 2. Weights sent to central server. 3. FedAvg aggregation completes.	1. All 3 clients trained. 2. Global model updated via FedAvg.	Pass	Federated training round completed without errors.
Model Prediction	Pneumonia Detection	Upload pneumonia X-ray image.	Output should show 'PNEUMONIA'.	Correct prediction shown.	Pass	Prediction accuracy aligned with model results.
Model Prediction	Normal Detection	Upload normal X-ray image.	Output should show 'NORMAL'.	Correct classification displayed.	Pass	Model correctly identifies normal cases.
FL Aggregation	FedAvg Weight Aggregation	1. Collect weights from 3 clients. 2. Apply FedAvg algorithm. 3. Update global model.	1. Weights averaged correctly. 2. Global model updated. 3. Model redistributed to clients.	1. FedAvg aggregation successful. 2. Global model accuracy improved.	Pass	Global model improved after each round.
Model Evaluation	Metrics Computation	1. Run model on test set. 2. Compute metrics. 3. Display results.	1. Accuracy, Precision, Recall, F1, AUROC computed. 2. Results displayed in table.	1. EfficientNet-B4: 90.71%. 2. All metrics computed correctly.	Pass	Metrics match expected values from training.
Privacy Preservation	No Raw Data Transfer	1. Run federated training. 2. Monitor data transfer between clients and server.	Only model weights transferred, no raw images.	Raw data remained at each hospital node.	Pass	Privacy constraint fully maintained.

Model Saving	Checkpoint After Each Round	1. Complete one federated round. 2. Save model checkpoint. 3. Verify file saved.	Model checkpoint saved to /kaggle/working/.	Checkpoint file found after each round.	Pass	Prevents loss of progress on disconnection.
Multi-Client FL	3 Hospital Node Simulation	1. Split training data into 3 parts. 2. Assign to 3 hospital clients.	All 3 clients train independently and send weights.	All 3 hospital nodes completed training successfully.	Pass	Federated simulation with 3 clients verified.
		3. Run training simultaneously.				

3.2.3 Architecture Document

The architecture in Sprint II focuses on model training, aggregation, and evaluation using federated learning. Each client trains the model locally using its own dataset, and the updated weights are sent to the central server.

The server aggregates the weights using the Federated Averaging (FedAvg) algorithm to create a global model. This process is repeated over multiple rounds to improve model performance.

The global model is then evaluated using test data. Performance metrics such as accuracy, precision, recall, and F1-score are calculated. Visualization tools like confusion matrix and ROC curve are used to analyze the results.

This architecture ensures efficient training, accurate evaluation, and privacy-preserving learning.

3.2.4 Outcome of Objectives / Result Analysis

The objectives of Sprint II were successfully achieved by training and evaluating the model using federated learning. The model showed good performance with high accuracy and recall.

Evaluation metrics and visualizations confirmed the effectiveness of the system. Overall, the sprint resulted in a reliable and well-performing pneumonia detection model.

3.2.5 Sprint Retrospective

Sprint II was successfully completed with all testing and evaluation objectives achieved. The federated learning model demonstrated good performance, achieving an accuracy of

approximately 88.94% with high recall in detecting pneumonia cases. The system was validated using standard evaluation metrics, and the results confirmed its reliability.

However, the training process required more time due to multiple communication rounds in federated learning, and a few false positives were observed. These challenges indicate the need for further optimization in model training and data handling.

The sprint provided useful insights into system performance and highlighted opportunities for improvement, such as reducing training time and enhancing model efficiency. Overall, the system is stable and ready for further enhancement and real- world application.

SPRINT II RETROSPECTIVE			
What went well	**What went poorly**	**What ideas do you have**	**How should we take action**
• Federated learning training completed successfully across multiple clients (hospitals). • The global model achieved high accuracy (~88.94%). • High recall achieved, ensuring effective detection of pneumonia cases. • Data preprocessing (resizing, normalization, augmentation) improved model performance. • Evaluation metrics (Accuracy, Precision, Recall, F1-score, AUROC) were computed successfully. • Visualizations (Confusion Matrix, ROC Curve) provided clear insights.	• Training time was relatively high due to multiple communication rounds. • Some false positives observed in the classification results. • Data distribution across clients was imbalanced. • Model complexity is high, requiring more computational resources. • Communication overhead between clients and server increased with more rounds.	• Use GPU acceleration and parallel processing to reduce training time. • Apply advanced data augmentation to improve generalization. • Balance data across clients or use weighted aggregation. • Experiment with lightweight models or pruning to reduce complexity. • Reduce communication rounds by using adaptive local epochs. • Implement early stopping based on validation performance.	• Optimize learning rate and batch size for faster convergence. • Apply class balancing techniques to reduce false positives. • Improve dataset distribution across clients. • Optimize model architecture (EfficientNet fine-tuning). • Use secure and efficient aggregation (FedAvg) with limited rounds. • Continuously monitor metrics and update the model iteratively.
Model training completed successfully	Training time was high	Use GPU optimization	Optimize learning rate & batch size
Global model accuracy ~88.94%	Some false positives	Improve data balance	Apply better augmentation
High recall for pneumonia detection	Model complexity high	Try lightweight models	Optimize architecture
Evaluation & visualizations worked well	Communication overhead high	Reduce communication rounds	Tune FL parameters

CHAPTER 4

RESULTS AND DISCUSSIONS

Project on IoT-based plant health and environmental monitoring system:

Project was successfully design as IoT-based plant health and environmental monitoring system which collects, transfer and visualize data in real-time and also analyze the environmental data and provide relevant suggestion to the user for maintaining the healthy plant environment and resources. Sensors are used to measure the environment such as temperature, humidity, soil moisture, air quality, rainfall, barometric pressure which continuously monitors by the ESP32 micro-controller. This system is also integrated with data visualization and analytical system where it interpret environmental condition.
VPD and various parameters were computed using sensors data in order to display the potential stress factors of plant and enable users to determine resource allocations such as water usage in proper irrigation management system.

4.1 PROJECT OUTCOMES (Performance Evaluation, Comparisons, Testing Results)

The proposed system utilizes Federated Learning with EfficientNet-B4 for pneumonia detection using chest X-ray images. The model was trained across multiple clients and evaluated using standard performance metrics. The system achieved a high accuracy of 90.71%, demonstrating its effectiveness in medical image classification**.**

4.1.1 CONFUSION MATRIX

The confusion matrix shows the classification performance of the model on the test dataset. The model correctly classified 184 normal and 382 pneumonia cases. It misclassified 50 normal images as pneumonia and only 8 pneumonia images as normal.

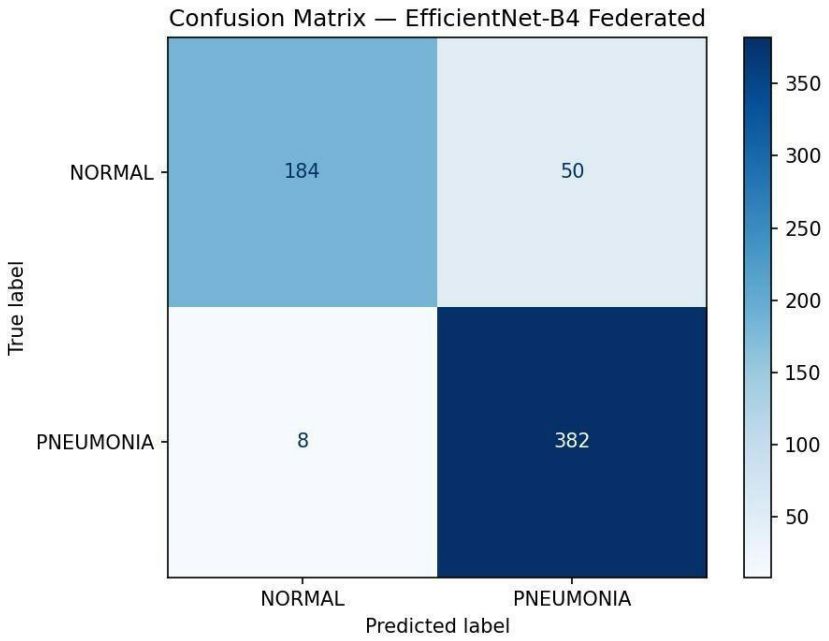


Figure 2: Confusion Matrix

The low number of false negatives indicates that the model is highly effective in detecting pneumonia cases, which is crucial in healthcare applications

4.1.2 FEDERATED TRAINING ACCURACY

The accuracy graph shows the performance across federated learning rounds. The model accuracy improved from 86.9% in the initial round to a peak of 90.7%.

Although minor fluctuations were observed, the final accuracy stabilized at around 88.9%, indicating successful learning and convergence of the federated model

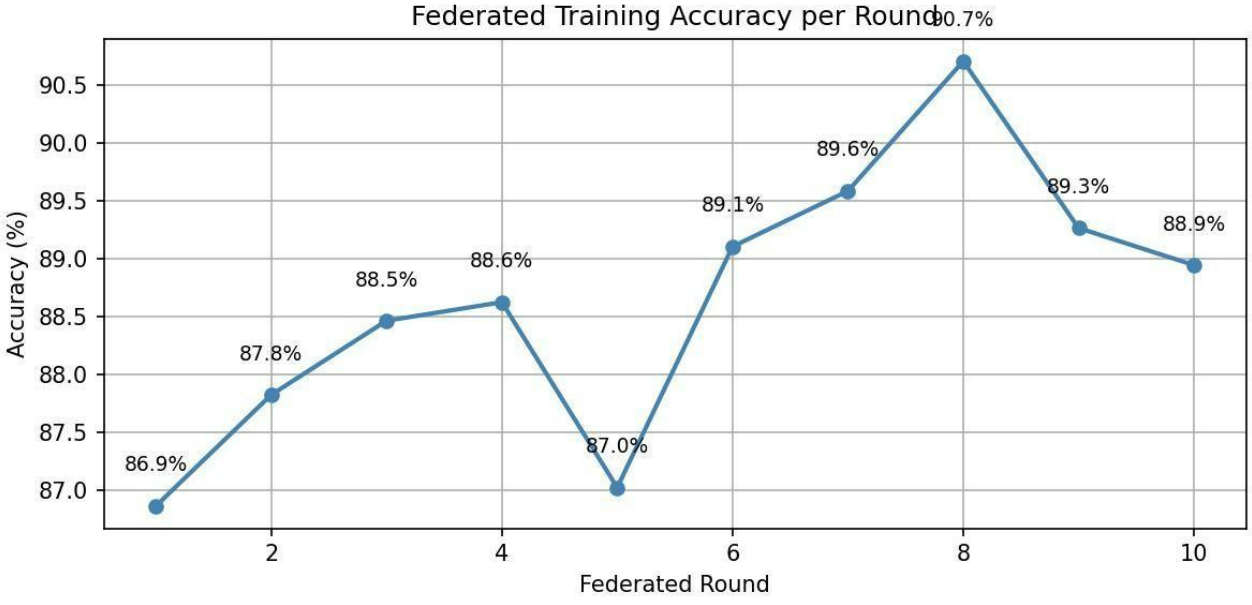


Figure 3:Federated Training Accuracy

4.1.3 ROC CURVE

The ROC curve demonstrates the classification ability of the model. The curve lies well above the diagonal, indicating strong performance.

The model achieved an AUC score of 0.964, confirming excellent ability to distinguish between normal and pneumonia cases.

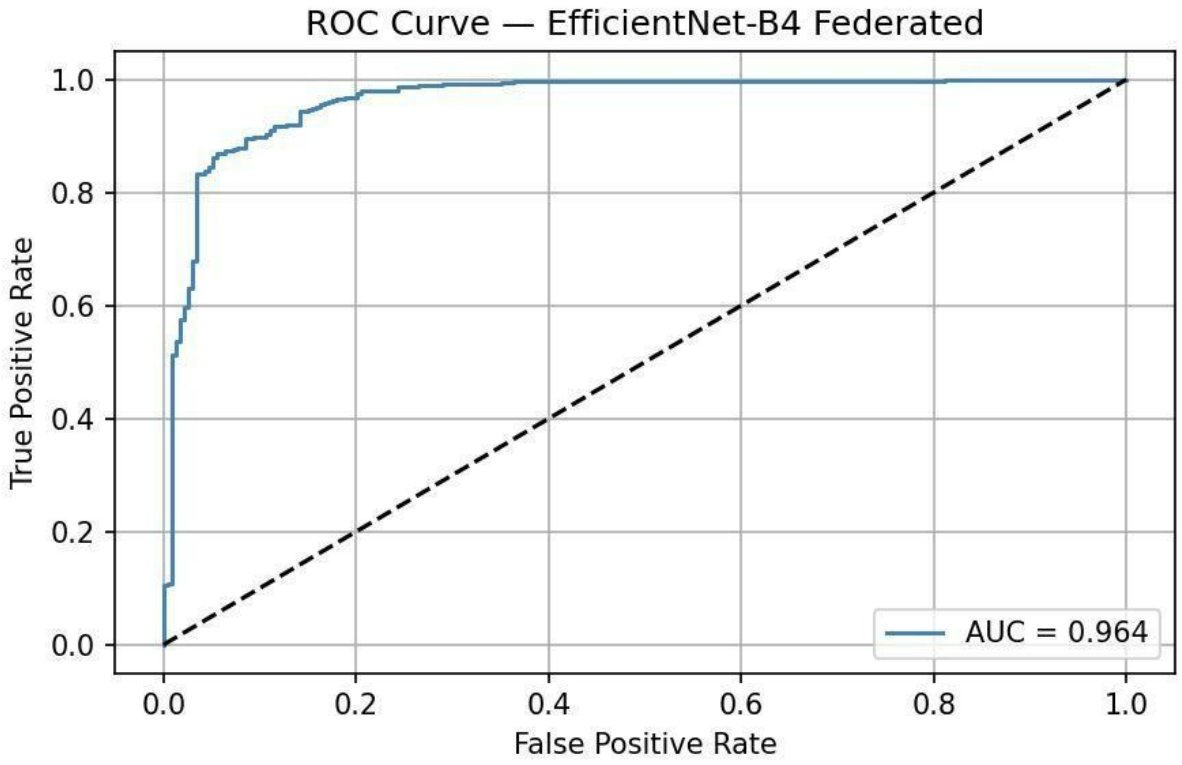


Figure 4: ROC Curve

4.1.4 PERFORMANCE COMPARISON OF MODELS

The performance of the proposed model was compared with other deep learning models such as DenseNet and ResNet. The results are shown in Table 4.1.

Model	Accuracy (%)	Precision	Recall	F1 Score	AUROC
EfficientNet-B4 (Federated)	**90.71**	**0.8843**	**0.9795**	**0.9294**	**0.964**
DenseNet	85.70	0.8100	0.9900	0.8900	0.970
ResNet	82.80	0.7800	0.9600	0.8600	0.960

The comparison shows that the proposed EfficientNet-B4 federated model outperforms the other models in terms of accuracy and overall performance. The higher recall and F1-score indicate better detection capability, especially for pneumonia cases.

4.1.5 RESULT ANALYSIS

Overall, the results confirm that the proposed system achieves high accuracy, strong recall, and excellent classification performance while maintaining data privacy through federated learning.